

Assessing the Quality of Pathotic Arguments

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Introduction

During argumentation appeals to emotion can be employed to persuade. This technique of persuasion is called **pathos**, and Arguments that employ **pathos** are **pathotic arguments**.

- How can we assess the quality and impact of pathotic arguments?
- Should we use pathotic arguments in scientific communication?

Goals

- Explore current approaches and limitations for assessing the quality of pathotic arguments.
- Evaluate existing datasets/corpora.

The outcome of this investigation could be used as a foundation for further research into social media, where pathotic arguments are common.

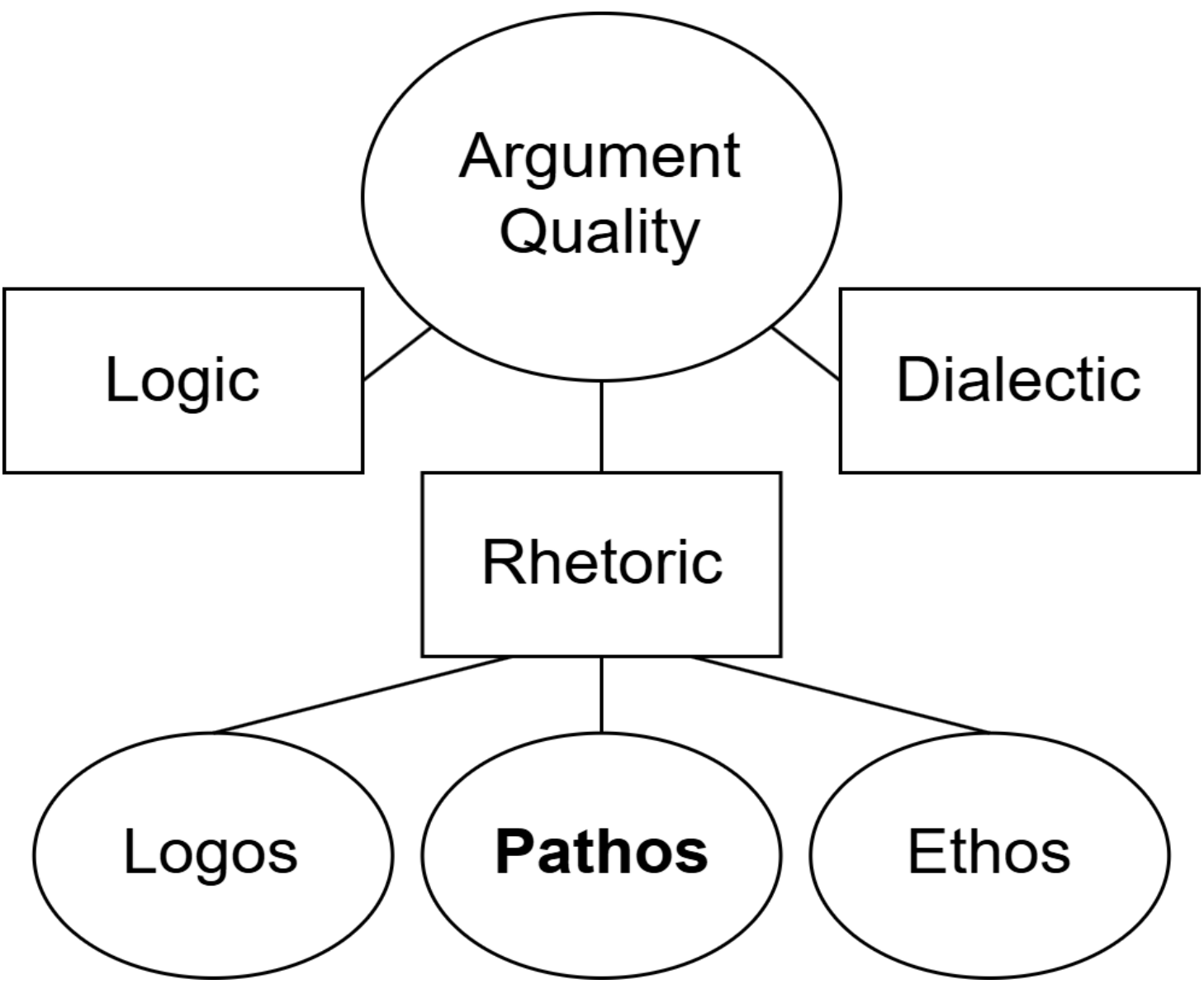
Pathos

- The arousal of emotions in an audience can be an effective form of persuasion:
- Using language that elicits emotion
 - Arguing that disagreeing/agreeing with the conclusion will lead to some kind of emotional state
 - Making certain argumentative moves such as concessions.

The nature of how this persuasion works is contentious, and the methods that constitute pathos are debatable.

Assessing Argument Quality

- A good argument has a higher argument quality than a bad argument
- Arguments can be annotated with a quality rating, and models trained to automatically assess argument quality
- Automatic assessment of the quality of pathotic arguments has not received much specific attention



An emotional debate about "Twilight"

Team Jacob

Bella should not have ended up with Edward, she should have ended up with Jacob

Emotion eliciting language: "ridiculous"

What happened in real life doesn't matter. You should consider what our friends will think about you when I tell them this!

Concessions/empathy

Team Edward

No that's ridiculous, Kristen Stewart and Robert Pattinson literally dated while filming Twilight

Argument from negative consequences

Well, I think we can all agree that she actually had the best chemistry with Carlisle anyway

Findings

- Some datasets are annotated for pairwise and individual argument quality (Habernal & Gurevych, 2016; Toledo et al., 2019) and to identify pathotic arguments (Gajewska et al., 2024). However, only a few datasets relate the two concepts:
- Carlile et al. (2018) annotate for quality and whether the argument is pathotic
 - Wachsmuth et al. (2017) annotate grades of 1-3 along various dimensions of argumentation including pathos
 - Konat et al. (2024) annotate emotion-eliciting language in the arguments and emotion-expressing language in reactions

Challenges

- Annotating argumentative text is already a hard task. Pathos adds an extra layer of subjectivity and annotator agreement is frequently low. Incorporating perspectivist metrics such as crowd truth (Aroyo & Welty, 2013), might yield interesting results.
- By making all annotators annotate each argument, and recording their motivations for each annotation, clearer insights can be gained into what makes annotating for pathos such a difficult task.

Directions for Future Work

- Create structured guidelines for pathotic quality
- Investigate related areas: computational humour, gestures, narratives
- Pathotic arguments in scientific communication

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